Introduction

The harmonic mean is the reciprocal of the arithmetic mean of reciprocals. It is appropriate when averaging rates, speeds, or ratios where the denominator is fixed – for example, when a traveller covers equal distances at different speeds and you want the average speed. In machine learning, the harmonic mean of precision and recall gives the F1 score.

Prerequisites

The reader should know the arithmetic mean and be comfortable taking reciprocals in R.

Theory

For positive values $x_1, \ldots, x_n$,

$$\mathrm{HM}(x) = \frac{n}{\sum 1/x_i}.$$

The harmonic mean is always less than or equal to the geometric mean, which is less than or equal to the arithmetic mean: $\mathrm{HM} \leq \mathrm{GM} \leq \mathrm{AM}$, with equality when all x_i are equal.

The F1 score in binary classification is the harmonic mean of precision and recall:

$$F_1 = \frac{2 \cdot P \cdot R}{P + R} = \frac{1}{\frac{1}{2}\left(\frac{1}{P} + \frac{1}{R}\right)} \cdot (\text{rescaled}).$$

Assumptions

All values must be strictly positive. Negative values or zeros make the harmonic mean undefined or infinite.

R Implementation

```r
library(psych)

speeds <- c(40, 60, 80)  # km/h over three equal distances
psych::harmonic.mean(speeds)
mean(speeds)                 # incorrect for average speed over equal distances

# F1 score
precision <- 0.80
recall    <- 0.60
f1 <- 2 * precision * recall / (precision + recall)
f1

# Harmonic mean is appropriate for average rates
n <- 3
hm_speed <- n / sum(1 / speeds)
hm_speed
```

Output & Results

```
[1] 55.38       # harmonic mean of speeds
[1] 60          # arithmetic mean (incorrect for equal-distance travel)
[1] 0.686       # F1 score with P=0.80, R=0.60
```

Travelling equal distances at 40, 60, 80 km/h gives an average speed of 55.4 km/h, not 60 km/h. The F1 of 0.69 splits the difference between P and R, penalising the lower value more than the arithmetic mean would.

Interpretation

Use the harmonic mean when averaging rates with a fixed numerator or denominator (e.g., average speed when distance is constant). Use the F1 form when balancing two normalised scores where both values matter.

Practical Tips

- Zero or negative inputs break the harmonic mean; guard against them.
- The harmonic mean weights small values more heavily; a single small observation drives the result down.
- In meta-analysis of rates per unit time, harmonic-mean sample size is sometimes used in effective-sample-size calculations.
- The F-beta score generalises F1: $F_\beta = (1 + \beta^2)PR/(\beta^2 P + R)$, with $\beta = 2$ emphasising recall and $\beta = 0.5$ emphasising precision.
- Reporting a harmonic mean without explanation is rare; specify why it is the appropriate summary in the methods.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Descriptive statistics and Table 1